Pole-compatible signed trace and a CM correspondence criterion

Signed block n/d at P=-U

Preserve x(jG), x(jQ), and the relative y sign

Six-coefficient norm det M(d+s*n0)

Keep e^0,e^1,e^2 and s*e^0,s*e^1,s*e^2; invert only units

Pole coefficient = [s]det / [e^2]det

First derivative of translated x suffices; origin chart handled separately

Divide by 4*y(U)^2, lift and subtract R0

Unique rational point; replay against the declared public point action

Derived construction, 2026-11-08, T-1922. The signed cyclic trace from Rounds56-57 now has a local evaluator at its rational double poles. A determinant in a six-coefficient nilpotent algebra extracts the marked coefficient without identifying the singular baby root in the extraction path. Native checks recover456 complete toy actions and32 selected full-width P-192 block actions. These are exact producer checks and written derivations, pending trusted review. [pcc:unverified]

The object being evaluated is C(P,R0)=sum_j x(P+jG)x(R0+jQ), for public Q=[k]G and a fixed extension-field point R0 with2R0 nonrational. For a nonzero rational U, put P0=-U and e=x(P)-x(P0). Only the jG=U summand has a pole, and its leading coefficient is4*y(U)^2*x(R0+[k]U). Thus the coefficient divided by4*y(U)^2 yields x(R0+[k]U); lift its two y choices, subtractR0 and retain the rational point. If both choices were rational then2R0 would be rational, a contradiction. This is a point-action interface; extracting a scalar subsequently still requires the orbit-representative construction. [pcc:unverified]

A block uses S1=P+A and S2=R0+B, where(A,B) is a signed giant, and babies Tj=(jG,jQ). Write h(T)=product_j(T-x(jG)), L=K[T]/h, A_b(x(jG))=x(jQ), B_b(x(jG))=y(jG)y(jQ), and K=F_(p^2). With u=x(S1),v=x(S2), H(T,u)=u*T^2+(a+u^2)*T+a*u+2b, the paired trace is n/d, where d=(T-u)^2*(A_b(T)-v)^2 and n=2*(H(T,u)*H(A_b(T),v)+4*y(S1)*y(S2)*B_b(T)). This retains the relative sign of the two elliptic factors. [pcc:unverified]

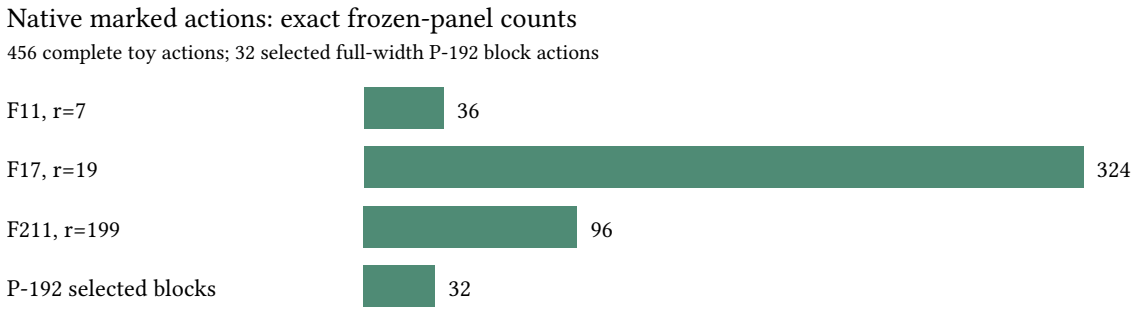
Coefficient theorem. Suppose exactly one component of the separable baby algebra has d_j(e)=c_j*e^2+O(e^3), with c_j nonzero, and every other d_l(0) is nonzero. In J=K[e,s]/(e^3,s^2), form R(e,s)=det(M_(d(e)+s*n(0))). Then [e^2]R(e,0)=c_j*product_(l!=j)d_l(0), whereas [s*e^0]R=n_j(0)*product_(l!=j)d_l(0). Their ratio is [e^-2]Tr_L(n/d). This follows by splitting the separable algebra and expanding the product of its factors; numerator derivatives do not enter either coefficient. When the constant determinant is nonzero, the block is regular and contributes zero. [pcc:unverified]

Sufficient precision. For finite S1(P0), translation invariance of dx/(2y) gives dx(S1)/dx(P)=y(S1(P0))/y(P0)=c. Use u(e)=u0+c*e in d, keeping S2 fixed. The omitted true second-order term changes the e^2 coefficient of d(T,e) by a multiple of(T-u0), which vanishes at the singular baby root. Therefore it cannot change the leading determinant coefficient or the extracted residue. This is a sufficient lift for those two coefficients, not a complete second-order point jet. The native control adds(T-u0) to the retained d2 and checks both coefficients unchanged at396 genuine poles, through each norm method. [pcc:unverified]

Charts and rank. If S1(P0)=O, both S1+/-Tj are finite because each baby is nonzero; the whole block is regular and is skipped. If P0=A, ordinary doubling gives S1=2A and the derivative ratio remains valid. The independent derivative checks expand the chord or tangent slope through first order. The second elliptic denominator never vanishes: R0 plus a rational point cannot equal a rational baby or its negative. Distinct baby x-coordinates imply at most one singular component, so the constant multiplication matrix has rank m-1 at a genuine pole. [pcc:unverified]

Algorithm and charged cost. Gaussian elimination uses full row/column pivoting and only inverts entries with nonzero constant coefficient. A unit c*(1+v) has inverse c^-1*(1-v+v^2-v^3), since v^4=0 in J. The final1x1 residual is multiplied without inversion; lower constant rank is an explicit failure. The implementation uses O(m^3) ring operations per giant and O(m^2) storage, plus baby encoding, matrix construction, explicit point inputs and tails. With2n giants the determinant cost is O(n*m^3). The six-coefficient ring has constant size. No fast resultant bound is assigned to this cubic backend. A future compressed evaluator must supply precisely these coefficients and pay for its representation and point batches. [pcc:unverified]

Frozen complete panel. The F11,r7 and F17,r19 controls cover every nonzero Q and U; the F211,r199 control covers twelve frozen Q labels and eight U labels. The outputs record456 actions,396 genuine block poles,60 explicit tail poles,108 origin chart skips and108 doubling translations, including24 genuine doubling poles. All396 reversed relative-sign residues differ from the correct residue. The determinant output and a product-over-roots norm calculation agree on every retained coefficient. They share field/curve helpers but the reference does not perform matrix elimination. [pcc:unverified]



P-192 binding. Use the standard full-width parameters and public Q in ../round52-p192/input.json, and R0=6S from ../round54-p192/P192-shift.json. Freeze babies1,3,5,7, giants16,32, both giant signs and both pole signs, giving32 selected block poles. Multiplication matrices are constructed by Vandermonde conjugation; their determinants match the root-product calculation. Every normalized coefficient decodes to the corresponding signed public multiple ofQ, with no private scalar supplied. Four origin translations and four regular doubling translations pass their chart checks. This evaluates selected blocks; it does not construct or evaluate the full P-192 trace. [pcc:unverified]

CM correspondence criterion, derived. Let E be ordinary with End(E_bar)=Z[pi], discriminant D=t^2-4p<0. Let C be an elliptic subgroup of E x E with nonconstant projections alpha,beta of degrees a,b, containing(G,Q). For theta=beta*dual(alpha), deg(theta)=a*b and theta(G)=[a]Q. Write theta=u+v*pi. Completing the norm gives4*a*b=(2u+t*v)^2+|D|*v^2. If4*a*b<|D| then v=0, u^2=a*b and [a]Q=[u]G. Thus a genuinely noninteger CM correspondence must pay this product-degree threshold. This is a necessary point-action constraint, not a sufficient certificate that each compatible triple definesC. [pcc:unverified]

P-192 consequence and boundary. The saved Round4 discriminant certificate makes D fundamental, hence End(E_bar)=Z[pi], conditional on the standard trace input recorded there. Here |D|=241093790603361101225441612331139756649492725178968653595 15. For a,b<=4096,4*a*b<=67108864<|D|, so an elliptic subgroup correspondence in that degree box must satisfy the small integer relation above. This does not bound arbitrary algebraic carriers, multicharts, arithmetic circuits or high-degree compressed maps such as1-pi. A small class number or L-value supplies neither the signed encoding nor a cheaper coefficient evaluator by itself. The next theory task is to relate the actual horizontal relation to a compressed carrier with an explicit point action, and test its degree/representation cost against this constraint. [pcc:unverified]

Verification and custody. The scope was committed to GitHub before the affected panel, and each added P-192 or precision control has a preserved before-data amendment/source publication. Fresh-copy native builds reproduce both panel outputs through both norm methods. Altering the trace numerator rejects with exit101; corrupting a retained coefficient also rejects with exit101. The required broad library check and all actual errors are recorded separately in validation.json. No cloud job or generator was started or modified. The scope record freezes exact proof checks, not an empirical performance experiment; no statistical or timing conclusion is drawn. [pcc:unverified]

Sources checked (KB unavailable). The KB MCP tools are not loaded and the migrated scripts/research-kb wrapper is absent. Stacks Project, https://stacks.math.columbia.edu/tag/0BIE , defines norm by the multiplication determinant. BFLS, https://arxiv.org/pdf/2003.10118 , Example4.13 explicitly permits jet algebras and section4.15 cautions that replacing the elliptic coordinate needs a suitable addition relation. MIT18.783 Lecture6, https://math.mit.edu/classes/18.783/2025/LectureNotes6.pdf , Theorem6.7 and Lemma6.10 supply the dual-isogeny identities. These standard ingredients underlie the construction and CM criterion; this round makes no prior-art novelty claim. [pcc:unverified]